
Intelligence Engineering

Solving Real World Problems At Scale

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LECTURE I

Overview of a Machine Learning System

1

The small part and the big picture

Don't confuse the model with the machine learning system.

Enabling the delivery of the logic

The **model** is the small part of a machine learning system. With foundation models and AI engineering, that part got smaller still. The essential work of a machine learning system remains and stacks around business understanding, infrastructure, design pattern and data engineering.

Before learning about solving a problem with machine learning, let's first assess when it is actually a good idea to do so.

(1) Learn

The problem you are solving, requires your system to have the capability to learn. Learning can look different depending on the problem and its context and boundary conditions. You already know about supervised learning, unsupervised learning and reinforcement learning paradigms.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

(2) Pattern

Learning requires the existence of a pattern, as opposed to noise or randomness. It is not trivial nor obvious whether there is a pattern in the data, and whether it is learnable. Complex pattern is more difficult to embed or state explicitly, this is where machine learning has an edge over classical approaches.

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(3) Data

Data is available, accessible or acquirable. The data is of sufficient quality and quantity to support learning, (and to validate for that matter). If your approach does not require data, it is because you build on embeddings or foundation models, which themselves are built on data in the first place, or you need to rely on data acquisition and curation as part of your system.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

(4) Predictions

Machine learning is about making predictions, always. Predictions can be of different types, such as classification, regression, ranking, clustering, generation and on and on. By reframing problems in terms of predictions, we can leverage machine learning to solve them. Predictions are approximations by nature and therefore come with uncertainty, which needs to be accounted for.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

(5) Unseen Data

In a closed system or domain, the data is finite and the system can be fully specified by optimization. However, often the domain is open and infinite, and the system needs to be able to generalize to unseen data. Of course, the unseen data is not completely unknown, it is drawn from the same distribution (i.i.d.) as the training data and is thus able to generalize to the unseen data inside a specific domain or distribution.

Machine learning is an approach to (1) learn (2) complex patterns from (3) existing data and use these patterns to make (4) predictions on (5) unseen data.

Catalysts

A catalyst is a substance that increases the rate of reaction without undergoing change itself.

The task is repetitive.

In a sense that we have a lot of data to learn a pattern from.

It scales.

The need to scale a specific prediction capability, not only a few times.

The context changes.

Meaning that hard coded or other solutions will be brittle and break.

Being wrong is not too bad.

Approximations on unseen data. Failures need to be manageable.

When Not to Use Machine Learning

It is unethical.

Well, that might also be a general rule. But doing stuff at scale, without human oversight, has its implications.*

There's a simple solution.

A simple solution, heuristic, or baseline should always be the first step regardless.

It's not economically viable.

The value of the prediction, needs to be greater than the cost of the system, including development, deployment and maintenance.

* More? Enroll to my Philosophy of AI course.

Describe two use-cases and the rationale for whether or not to use machine learning.

One Machine Learning Use-Case
and **One** which is not.

Mapping the value, 2018

$P(y|x)$

The map: value in the trillions

McKinsey sized **400+** use cases across **19** industries and nine business functions: deep learning promised **\$3.5 to \$5.8 trillion** in value per year, the largest shares in marketing and sales plus supply chain and manufacturing.

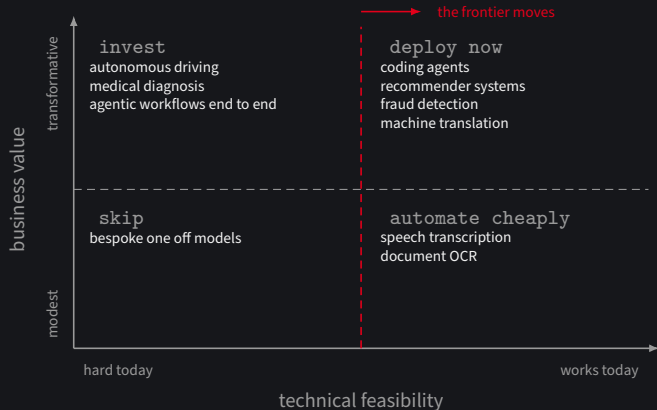
The ground: months per model

Algorithmia surveyed **745** practitioners in October 2019: most companies needed **8 to 90 days** to deploy a single model, 18% needed longer than 90 days, and deployment time grew year over year.

The value was mapped in trillions while shipping one model took a quarter: feasibility, not value, was the bottleneck.

[1, 2]

Value over technical feasibility



SWE-bench Verified rose from 60% to near 100% within a year.

Claude Opus 4.1 wins or ties against human experts on 47.6% of GDPval tasks.

Among organizations above \$1 billion revenue, 40% now scale AI agents, up from 27% in one year.

The field is moving, 2026

Adoption is near universal

- **88%** of organizations use AI in at least one business function, **72%** use gen AI.
- Gen AI reached 53% population adoption within three years, faster than the PC or the internet.
- US private AI investment reached \$285.9 billion in 2025.

Value capture is not

- **80%** of AI users report individual productivity gains.
- **37%** of organizations attribute any EBIT impact to AI, flat since 2025.
- **6%** qualify as AI high performers.

Adoption climbs, value capture flat

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What the labs measure

OpenAI: GDPval

1,320 tasks from **44** occupations across nine sectors, graded head to head by experts averaging 14 years in the field. Frontier models finish them about **100×** faster and **100×** cheaper than the experts.

Anthropic: Economic Index

49% of jobs show Claude on at least a quarter of their tasks. Augmentation leads automation **52% to 45%**. The single most common task: modifying software to correct errors, 6% of all usage.

Broad across occupations, thin within them, and the densest use is code: the labs' own traffic redraws the feasibility axis every quarter.

2

Understanding Machine Learning Systems

Research and Production are not the same.

Horizontal versus Vertical

Nothing motivates **system** thinking more than the need to go **horizontal**. Academia is vertical, with a narrow focus on a single problem, where drilling deep, fast is rewarded and focus allows to teach a wide range of concepts.

	Research	Production
Objectives	State of the Art highly engineered models	Stakeholder outcome driven
Compute	Training and high throughput	Fast inference with low latency
Data	Often static benchmark oriented	Shifting domains and structures
Usability metrics	Usually not considered	UI, Fairness, Robust, Interpretability

Stakeholders and Objectives

Tinder for horses. In research the objective is one number on one benchmark. Ship the same model as a product and five people want five different things from it, each arriving with a solution space already in mind: **conflicting requirements** are the norm, and not all of them are must haves.

ML engineers

Objective better matches.
Solution space more data, a bigger model.

Platform team

Objective nothing pages at 3am.
Solution space freeze the model, build the platform.

Product team

Objective the swipe feels instant.
Solution space a latency budget of 100 ms.

Sales team

Objective revenue per match.
Solution space rank the premium stallions first.

Manager

Objective margin.
Solution space fewer models, and maybe fewer engineers.

Eisenhower: objective over solution space

[7]

Leaderboards and Benchmarks

A leaderboard scores the **model**. A system is not a model.

- What a benchmark measures is one metric on one frozen test set. What it never measures is the pipeline around it: acquisition, cleaning, labelling, feature engineering, deployment, monitoring.
- Thousands of entries against the same test set is multiple hypothesis testing, so part of the podium is pure chance. Scanning 10,000 seeds on CIFAR 10 spreads accuracy from 89.01 to 90.83 percent, 1.82 points, with nothing changed but the seed.
- A benchmark is a proxy for a single metric. It was never forged in production, where objectives conflict, data shifts, and the users are real.

Trust me bro benchmark: no axis, no error bars

[3]

Computational priorities

Research: fast training

- Training is the bottleneck: many models, each doing multiple passes over the data.
- Prioritizes **high throughput**, how many queries are processed in a period of time.

Production: fast inference

- Once deployed, inference is the bottleneck: the model's job is to generate predictions.
- Prioritizes **low latency**, the time from receiving a query to returning the result.

Processing	Average latency	Throughput
One query at a time	10 ms	100 queries/second
One query at a time	100 ms	10 queries/second
Batches of 10 queries	10 ms	1,000 queries/second
Batches of 50 queries	20 ms	2,500 queries/second

Batched, higher latency might also mean higher throughput.

Latency is a distribution

Ten requests, latency in ms: 100, 102, 100, 100, 99, 104, 110, 90, **3,000**, 95.

The average latency is 390 ms, which makes your system seem slower than it actually is.

Better to think in percentiles: p50 is the median, p90 and p99 expose the outliers.

The p90 here is 3,000 ms, the one request to investigate.

Latency matters a lot.

2017, Akamai: a 100 ms delay can hurt conversion rates by 7%.

2019, Booking.com: about 30% more latency cost about 0.5% in conversion rates, “a relevant cost for our business”.

2016, Google: more than half of mobile users leave a page that takes more than three seconds to load.

On Amazon the slowest requests come from the most valuable customers.

Research: static

- Clean and well formatted
- Static, so the community can benchmark
- Quirks of the dataset are known
- Open source scripts to feed your models
- Historical data, already stored somewhere

Production: constantly shifting

- Messy, noisy, possibly unstructured
- Likely biased, and you do not know how
- Labels sparse, imbalanced, or incorrect
- Changing requirements can mean relabeling
- Privacy and regulatory concerns
- Generated by users, systems, third parties

Andrej Karpathy, director of AI at Tesla, drew the data problems of his PhD against those of his time at Tesla.

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Fairness

During the research phase, a model is not yet used on people, so it is easy for researchers to put off fairness as an afterthought.

“Let’s try to get state of the art first and worry about fairness when we get to production.”

When it gets to production, it is too late.

Optimize for better accuracy or lower latency and you can show that you beat state of the art.

As of writing the book, there is no equivalent state of the art for fairness metrics.

Fairness at scale

Machine learning algorithms do not predict the future, they encode the past, thus perpetuating the biases in the data.

- Your loan application might be rejected because the algorithm picks on your **zip code**, which embodies biases about one's socioeconomic background.
- Your resume might be ranked lower because the ranking system employers use picks on the **spelling of your name**.
- Your mortgage might get a higher interest rate because it relies partially on **credit scores**, which favor the rich and punish the poor.

Deployed at scale, they discriminate at scale, judging millions in split seconds.

In 2019, Berkeley researchers found that both face-to-face and online lenders rejected **1.3 million** creditworthy Black and Latino applicants between 2008 and 2015. With the income and credit scores kept and the race identifiers deleted, the mortgage application was accepted.

McKinsey & Company, 2019: only **13%** of the large companies surveyed said they are taking steps to mitigate risks to equity and fairness.

Interpretability

In early 2020, Turing Award winner Geoffrey Hinton posed a heatedly debated question. Suppose you have cancer and you have to choose.

Black box AI surgeon

90% cure rate, and it cannot explain how it works.

Human surgeon

80% cure rate.

“Do you want the AI surgeon to be illegal?”

Put to a group of 30 technology executives at public nontech companies, half of them wanted the AI surgeon to operate on them. The other half wanted the human surgeon.

Interpretability in production

Most machine learning research is still evaluated on a single objective, model performance, so researchers are not incentivized to work on model interpretability. In industry, interpretability is not just optional for most use cases but a requirement.

- Important for **users**, both business leaders and end users, to understand why a decision is made, so that they can trust a model and detect potential biases.
- Important for **developers**, to be able to debug and improve a model.

Most of us are comfortable using a microwave without understanding how it works. Many do not feel the same way about AI, especially if that AI makes important decisions about their lives.

As of 2019, only **19%** of large companies are working to improve the explainability of their algorithms.

Discussion

$P(y|x)$

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Machine Learning Systems Versus Traditional Software

$P(y|x)$

Traditional software engineering

- An underlying assumption that **code and data are separated**.
- Keep things as modular and separate as possible: separation of concerns.
- Many of its tools do work for machine learning.

Machine learning systems

- **Part code, part data**, and part artifacts created from the two.
- Applications with the most and best data win: companies improve data, not algorithms.
- Data can change quickly, so applications need to be adaptive.

ML production would be a much better place if ML experts were better software engineers.

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Test and version the data too

In traditional software engineering, you only need to focus on testing and versioning your code. With machine learning, we have to test and version our data too, and **that is the hard part**.

- How to version large datasets?
- How to know if a data sample is good or bad for your system?

Not all data samples are equal. Some are more valuable to your model than others.

Indiscriminately accepting all available data might hurt performance and make your model susceptible to data poisoning attacks.

Your model has already trained on **one million** scans of normal lungs and only **one thousand** scans of cancerous lungs.

A scan of a cancerous lung is much more valuable than a scan of a normal lung.

Size, speed and visibility

Size

As of 2022 it is common for models to have **hundreds of millions, if not billions, of parameters**, needing gigabytes of RAM.

Speed

Getting large models into production, **especially on edge devices**, is a massive challenge, as is running them fast enough.

Visibility

Monitoring and debugging in production is also nontrivial: models get complex, with a **lack of visibility** into their work.

An autocompletion model is useless if the time it takes to suggest the next character is longer than the time it takes for you to type.

The good news: these challenges are being tackled at a breakneck pace

In 2018 people said BERT was too big, too complex and too slow to be practical. The pretrained large BERT model has **340 million** parameters and is **1.35 GB**. Two years later, BERT and its variants were used in almost every English search on Google.

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